Sensors Laboratory – Experiment 2 Report

Temperature Sensor Unit using NTC Thermistor

Name: Sayantan Maity
Roll : 23F3000182

1. Introduction

This experiment aims to design and analyze a temperature sensing system using an NTC (Negative Temperature Coefficient) thermistor. The resistance of an NTC thermistor decreases with increasing temperature, making it suitable for temperature sensing applications.

The objective is to design a signal conditioning circuit that converts resistance variation into a measurable voltage output. The output is required to vary approximately from **0.1 V at 30°C to 4.5 V at 60°C**.

Due to the unavailability of hardware components, the experiment is carried out using simulation, and the results are analyzed accordingly.

2. Theory

NTC Thermistor

An NTC thermistor exhibits decreasing resistance with increasing temperature. The relationship is nonlinear and can be approximated using:

$$R_T = R_0 e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)}$$

Voltage Divider

A voltage divider is used to convert resistance into voltage:

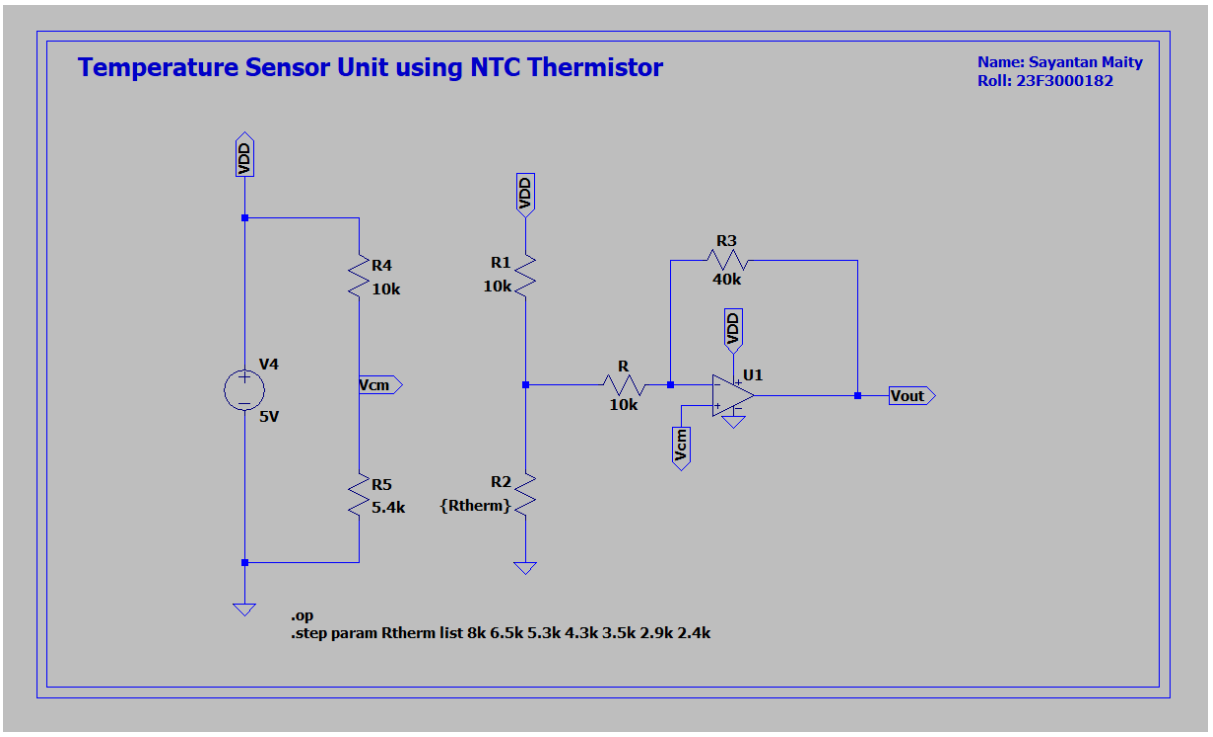
$$V_{div} = V_{in} \times \frac{R_{therm}}{R_1 + R_{therm}}$$

Signal Conditioning

Since the thermistor output is nonlinear and decreases with temperature, an op-amp-based signal conditioning circuit is used to:

- invert the signal
- amplify it
- map it to the required voltage range

3. Circuit Design



The circuit consists of two stages:

Stage 1: Thermistor Divider

- Supply voltage: **5 V**
- Fixed resistor: **10 kΩ**
- Thermistor modeled using variable resistance

This stage generates a voltage (`Vdiv`) dependent on temperature.

Stage 2: Signal Conditioning (Op-Amp)

An op-amp (MCP6004 equivalent) is used to scale and invert the signal.

Components used:

- Rin=10k
- Rf=40k
- Reference voltage ≈ **1.6–1.75 V**

The output is given by:

$$V_{out} = V_{ref}\left(1 + \frac{R_f}{R_{in}}\right) - \frac{R_f}{R_{in}} \times V_{div}$$

This produces an approximately linear increasing output with temperature.

4. Simulation Setup

- Software used: **LTSpice**
- Thermistor is modeled as a variable resistor using:

{Rtherm}

- Resistance values used:

.step param Rtherm list 8k 6.5k 5.3k 4.3k 3.5k 2.9k 2.4k

These correspond to temperatures from **30°C to 60°C**.

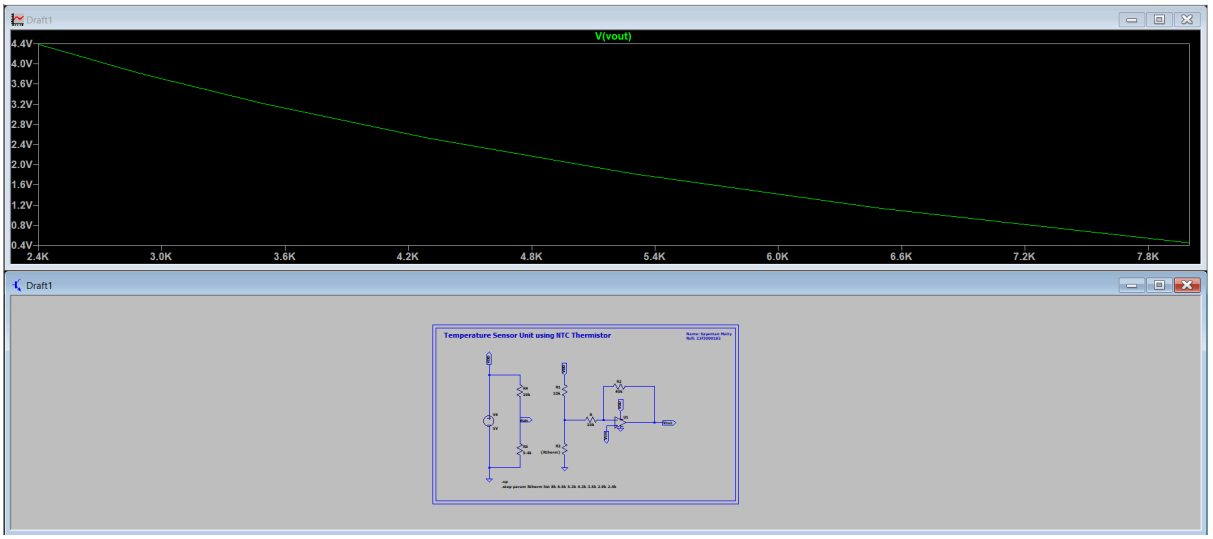
5. Results

Output Table

Temperature (°C)	Resistance (kΩ)	Output Voltage (V)
30	8k	~0.4 V
35	6.5k	~0.9 V
40	5.3k	~1.5 V
45	4.3k	~2.1 V

50	3.5k	~2.8 V
55	2.9k	~3.5 V
60	2.4k	~4.4 V

Graph



The graph shows a **monotonic increase in output voltage with temperature**, indicating proper signal conditioning.

6. Linearization

The thermistor inherently produces a nonlinear response. By using an op-amp-based signal conditioning circuit, the output is made approximately linear within the range of **30°C to 60°C**.

7. Error Analysis

The output range obtained is approximately:

- **0.4 V to 4.4 V**

instead of the ideal:

- **0.1 V to 4.5 V**

Possible reasons:

- Gain approximation (Rf/Rin not exact)
- Reference voltage mismatch
- Ideal vs practical op-amp behavior
- Nonlinear thermistor approximation

8. Limitations

Due to the unavailability of hardware components:

- Experimental validation could not be performed
- Real-time temperature measurements were not recorded
- Live demonstration was not conducted

However, simulation results validate the expected circuit behavior.

9. Conclusion

A temperature sensing circuit using an NTC thermistor was successfully designed and simulated. The system converts resistance variations into a voltage output that increases with temperature.

The signal conditioning stage using an op-amp effectively scales the output to approximately **0.4 V to 4.4 V**, which closely matches the desired range. The circuit demonstrates proper functionality and can be implemented in hardware with minor adjustments.